

Phase I/II Operating Characteristics

TITE versus new non-TITE: one-stage, two-stage highest utility, and two-stage top-2 randomized

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
TITE one-stage	19.1	8.4	1.9	0.3	0.1	70.2
New non-TITE one-stage	17	7.2	1.7	0.1	0.1	73.9
TITE 2-stage highest utility	19.4	7.7	1.2	0.3	0	71.4
New non-TITE 2-stage highest utility	18	7.2	1.2	0	0.2	73.4
TITE 2-stage top-2 randomized	18.7	8.7	3.9	0.7	0.2	67.8
New non-TITE 2-stage top-2 randomized	17.4	7.2	2	0.8	0.1	72.5
Mean administrations by dose						
TITE one-stage	11.6	4.7	2	0.9	0.3	
New non-TITE one-stage	11.1	4.4	2.3	0.9	0.3	
TITE 2-stage highest utility	11.5	4.6	2.1	0.9	0.3	
New non-TITE 2-stage highest utility	11.1	4.4	2.3	1	0.3	
TITE 2-stage top-2 randomized	11.1	4.5	2	0.8	0.3	
New non-TITE 2-stage top-2 randomized	11	4.4	2.3	1	0.3	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			711.6			
New non-TITE one-stage			672.8			
TITE 2-stage highest utility			710.3			
New non-TITE 2-stage highest utility			677.6			
TITE 2-stage top-2 randomized			690.1			
New non-TITE 2-stage top-2 randomized			677.8			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
TITE one-stage	30.1	22.1	20.7	15	12	0.1
New non-TITE one-stage	29.3	21.3	20.2	18.8	10.4	0
TITE 2-stage highest utility	32.6	23	20.1	14.2	10	0.1
New non-TITE 2-stage highest utility	34.1	21.3	16.6	16.4	11.5	0.1
TITE 2-stage top-2 randomized	28.1	24.9	19.1	17.5	10.3	0.1
New non-TITE 2-stage top-2 randomized	33.9	22.2	16.5	15.4	12	0
Mean administrations by dose						
TITE one-stage	7.1	6.7	5.9	5.7	4.6	
New non-TITE one-stage	7.3	6.7	5.7	5.7	4.6	
TITE 2-stage highest utility	7.4	6.9	5.9	5.5	4.4	
New non-TITE 2-stage highest utility	7.6	6.8	5.7	5.5	4.4	
TITE 2-stage top-2 randomized	6.1	6.1	5.6	5.7	5.3	
New non-TITE 2-stage top-2 randomized	6.6	6.1	5.6	6.1	5.6	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1082.9			
New non-TITE one-stage			999.5			
TITE 2-stage highest utility			1079.7			
New non-TITE 2-stage highest utility			1000.3			
TITE 2-stage top-2 randomized			1070.7			
New non-TITE 2-stage top-2 randomized			1005.9			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
TITE one-stage	11.2	45.4	27.3	11.9	1.6	2.6
New non-TITE one-stage	14.4	43.9	25.7	10.8	2.4	2.8
TITE 2-stage highest utility	12.3	49.8	25.6	8	1.4	2.9
New non-TITE 2-stage highest utility	14.2	47.4	25.8	7.7	1.7	3.2
TITE 2-stage top-2 randomized	8.4	49.3	25.4	12.7	1.6	2.6
New non-TITE 2-stage top-2 randomized	10.4	47.7	25.6	10.6	2.7	3
Mean administrations by dose						
TITE one-stage	5.2	11	7.7	4.2	1.4	
New non-TITE one-stage	5.2	11	7.6	4.2	1.4	
TITE 2-stage highest utility	5.3	11.1	8.1	3.7	1.4	
New non-TITE 2-stage highest utility	5.3	11.2	7.9	3.8	1.3	
TITE 2-stage top-2 randomized	5.7	8.6	7.2	4.2	1.7	
New non-TITE 2-stage top-2 randomized	5.7	9.5	8	4.6	1.8	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1100.7			
New non-TITE one-stage			1036.1			
TITE 2-stage highest utility			1102.4			
New non-TITE 2-stage highest utility			1035.7			
TITE 2-stage top-2 randomized			1088.9			
New non-TITE 2-stage top-2 randomized			1043.8			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
TITE one-stage	0.9	1.1	4.5	60	33.5	0
New non-TITE one-stage	1.4	0.9	5	58.6	34.1	0
TITE 2-stage highest utility	1	1.5	4.8	56.3	36.4	0
New non-TITE 2-stage highest utility	2	1.5	6.2	55.5	34.8	0
TITE 2-stage top-2 randomized	0.8	1.1	4.1	54.3	39.7	0
New non-TITE 2-stage top-2 randomized	0.4	0.8	3.6	56.6	38.6	0
Mean administrations by dose						
TITE one-stage	3.4	3.5	4.5	12.2	6.5	
New non-TITE one-stage	3.4	3.5	4.4	12.2	6.5	
TITE 2-stage highest utility	3.4	3.5	4.5	10.6	8	
New non-TITE 2-stage highest utility	3.4	3.5	4.4	10.6	8.1	
TITE 2-stage top-2 randomized	3.5	3.7	4.9	9.1	8.4	
New non-TITE 2-stage top-2 randomized	3.6	3.7	4.9	9.4	8.4	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1202.6			
New non-TITE one-stage			1079.9			
TITE 2-stage highest utility			1200.9			
New non-TITE 2-stage highest utility			1083.3			
TITE 2-stage top-2 randomized			1183			
New non-TITE 2-stage top-2 randomized			1076.6			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
TITE one-stage	13.1	25.4	45	10.9	0.6	5
New non-TITE one-stage	16.4	22.3	43.7	10.9	1.1	5.6
TITE 2-stage highest utility	15.8	28.1	43	6.8	0.4	5.9
New non-TITE 2-stage highest utility	16.3	23.9	44	8.4	0.9	6.5
TITE 2-stage top-2 randomized	11.8	25.9	42.7	13.6	1.1	4.9
New non-TITE 2-stage top-2 randomized	10.5	23.6	46.6	12.5	1.1	5.7
Mean administrations by dose						
TITE one-stage	7.1	8.2	9.4	3.8	0.6	
New non-TITE one-stage	7.1	7.9	9.8	3.6	0.7	
TITE 2-stage highest utility	7.7	8.2	9	3.3	0.9	
New non-TITE 2-stage highest utility	7.7	7.9	9.4	3.2	0.9	
TITE 2-stage top-2 randomized	7.1	7.6	7.4	3.6	0.9	
New non-TITE 2-stage top-2 randomized	7.4	8.2	8.4	4	1.1	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1124.4			
New non-TITE one-stage			1063.7			
TITE 2-stage highest utility			1117.6			
New non-TITE 2-stage highest utility			1065.2			
TITE 2-stage top-2 randomized			1083.5			
New non-TITE 2-stage top-2 randomized			1070.9			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
TITE one-stage	14.5	9.2	7.5	8.3	59.2	1.3
New non-TITE one-stage	14	7.9	6	7.8	63.2	1.1
TITE 2-stage highest utility	16.5	10.2	9	9.2	53.6	1.5
New non-TITE 2-stage highest utility	17.9	10	7.3	7.5	56	1.3
TITE 2-stage top-2 randomized	11.1	9.4	8.9	9.9	59.6	1.1
New non-TITE 2-stage top-2 randomized	11.7	6	5.5	8.5	67.3	1
Mean administrations by dose						
TITE one-stage	4.7	4.9	5	5.4	9.9	
New non-TITE one-stage	4.5	4.5	5	5.3	10.4	
TITE 2-stage highest utility	4.9	5	5.5	5.4	9.1	
New non-TITE 2-stage highest utility	4.8	4.7	5.3	5.4	9.6	
TITE 2-stage top-2 randomized	4.6	4.8	5.1	6.1	7.1	
New non-TITE 2-stage top-2 randomized	4.7	4.8	5.3	6.9	8.2	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1133.8			
New non-TITE one-stage			1040.1			
TITE 2-stage highest utility			1123			
New non-TITE 2-stage highest utility			1034.1			
TITE 2-stage top-2 randomized			1067.3			
New non-TITE 2-stage top-2 randomized			1021.7			